

**JKBOSE 2026**

**CLASS 11<sup>TH</sup>**

**PERMUTATION AND  
COMBINATIONS**

*Pnc.*

**MATHS**

**LECTURE-01**

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# TOPICS TO BE COVERED

**1**

**Fundamental Principle of Counting**

**2**

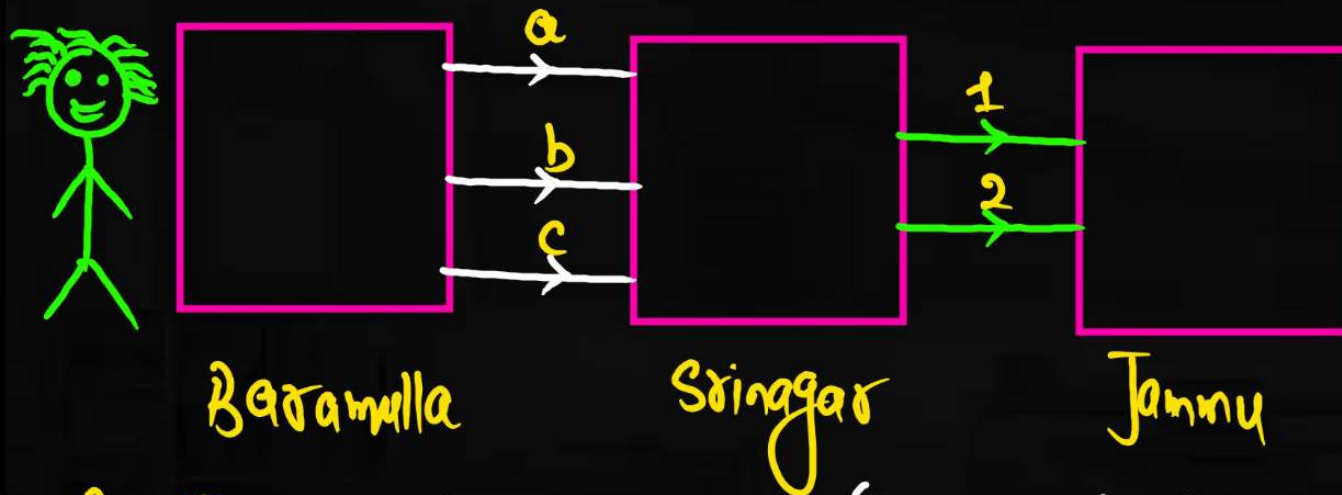
**Exercise Discussion 6.1**

**3**

**Related Problems**

# INTRODUCTION

## Fundamental Principle of Counting



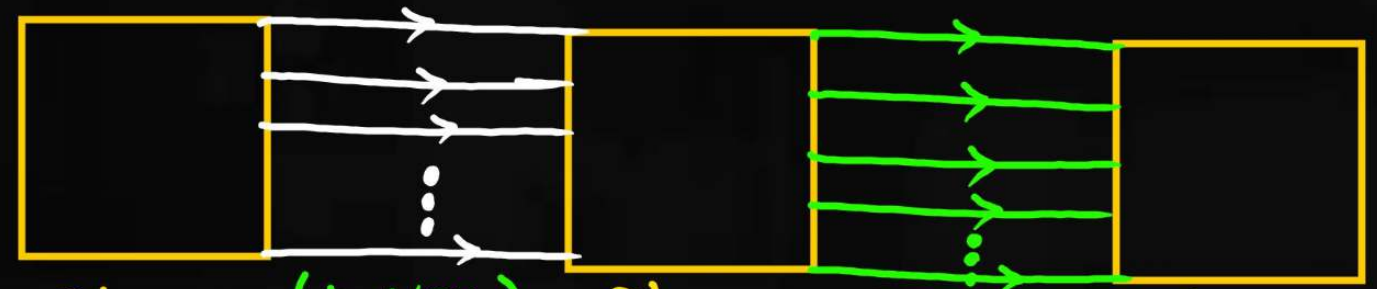
Question: In how many ways she (Akshade) can go from Baramulla to Jammu?

Ans) Total no. of ways =  $3 \times 2 = 6$

List of ways:

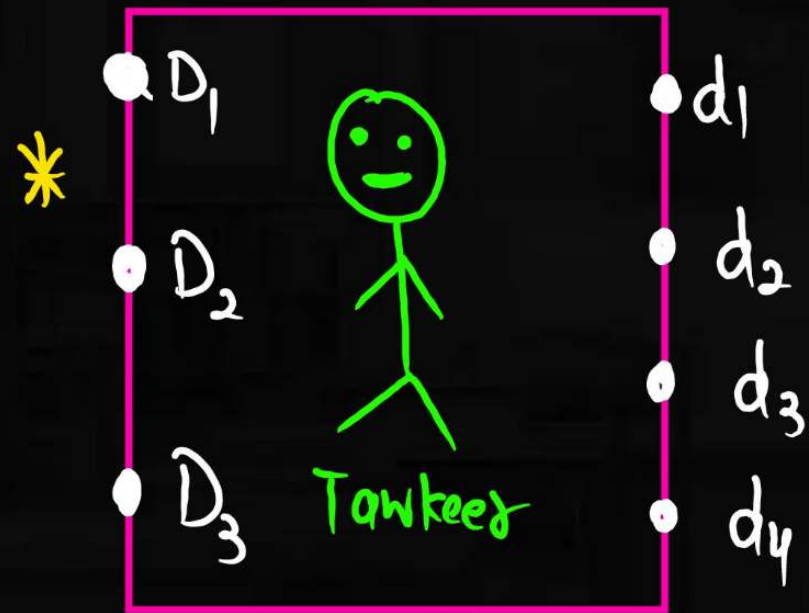
$a_1, a_2, b_1, b_2, c_1, c_2$ .

So, the required numbers = 6

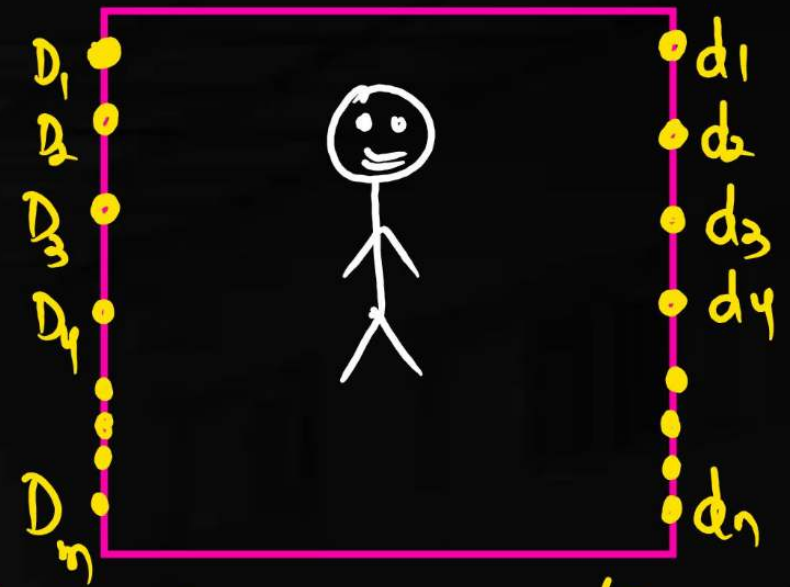


Place I (m ways) Place II (n ways) Place III  
 Here, the number of ways in which one can go from Place I to Place III =  $m \times n$

\* If an operation can be performed in  $m$  ways  
& another operation can be performed in  $n$  ways  
, then the number of ways of performing both the operations is  $m \times n$  (Multiplication rule)



Question) In how many ways can Tawkeer go outside the room?  
Ans) No of ways =  $3 + 4 = 7$



( $m$  ways)

( $n$  ways)

Total no. of ways =  $m + n$ .

\* If an operation can be performed in  $m$  ways and another operation can be performed in  $n$  ways, then the number of ways of happening exactly one operation at a time is  $m + n$  (Addition rule)

**EXAMPLE**

In a Class, there are 15 boys and 10 girls. How many ways a teacher can select 1 boy and 1 girl to represent the class at a Seminar.

| Boy | Girl |
|-----|------|
| 15  | 10   |

(Sol) Number of ways of selecting a Boy = 15

Number of ways of selecting a Girl = 10

Therefore by the fundamental principle of Counting, we get

Required Number of ways =  $15 \times 10 = 150$

Hence the result.

## EXAMPLE

Find the number of 4 letter words, with or without meaning, which can be formed out of the letters of the word ROSE, where the repetition of the letters is not allowed.

(Sol) By Fundamental principle of Counting, we get,

|   |   |   |   |
|---|---|---|---|
| 4 | 3 | 2 | 1 |
|---|---|---|---|

$$\text{Required No. of ways} = 4 \times 3 \times 2 \times 1$$

$$= 24$$

Hence the result.

R, O, S, E.

# EXAMPLE

If  $x < 4 < y$  and  $x, y \in \{1, 2, 3, \dots, 10\}$ , then find the number of ordered pairs  $(x, y)$ .

(Sol) By Fundamental principle of Counting;

Required number of ordered pairs

$$= 3 \times 6$$

$$= 18$$

Hence the result.

$$x, y \in \{1, 2, 3, 4, \dots, 10\}$$

$$x = 1, 2, 3$$

$$y = 5, 6, 7, 8, 9, 10$$

Question:



(a) 9 (b) 16 (c) 20 (d) 15.

$$\text{No. of ways} = 4 \times 5 = 20.$$

**QUESTION**

A college offers 6 courses in the morning and 4 in the evening. The possible number of choices with the student if he wants to study one course in the morning and one in the evening is-

☒ 24

☐ 2

☐ 12

☐ 10

(Sol) Required No. of ways =  $6 \times 4$   
 $= 24.$

**QUESTION**

A college offers 6 courses in the morning and 4 in the evening. The number of ways a student can select exactly one course either in the morning or in the evening.

**1**

24

**2**

6

**3**

4

**4**

10

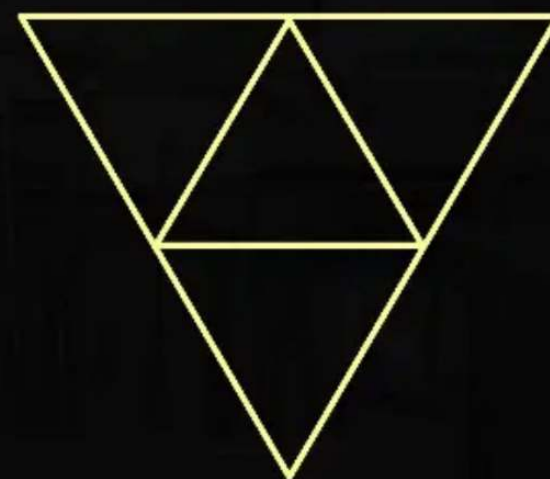
(Sol) Required number of ways  
 $= 6 + 4$   
 $= 10.$

## EXAMPLE

In how many ways can this diagram be coloured subject to the following two conditions?

- (i) Each of the smaller triangle is to be painted with one of three colours: red, blue or green.
- (ii) No two adjacent regions have the same colour.

*(Homework Problem)*



## QUESTION (NCERT EXAMPLAR)

There are four bus routes between A and B, and three bus routes between B and C. A man can travel round-trip in number of ways by bus from A to C via B. If he does not want to use a bus route more than once, in how many ways can he make round trip?

☒ 1

72

☐ 2

144

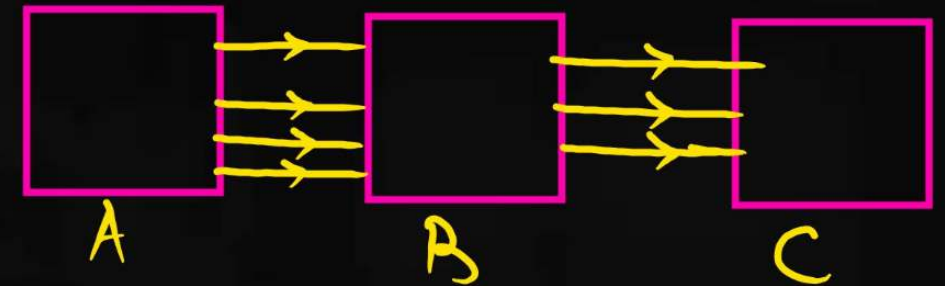
☐ 3

14

☐ 4

19

(Sol) Required Number of ways  
 $= (4 \times 3) \times (2 \times 3)$   
 $= 12 \times 6$   
 $= 72$



# QUESTION (NCERT EXAMPLAR)

In an examination there are three multiple choice questions and each question has 4 choices. Number of ways in which a student can fail to get all answer correct is

☒ 1 11

☒ 2 12

☒ 3 27

☒ 4 63

Q1 Q2 Q3

a a a

b b b

c c c

d d d

(4 ways) (4 ways) (4 ways)

1 1 1

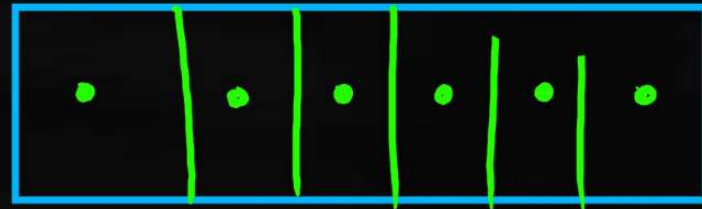
(Sol) Number of ways of attempting 3 Questions =  $4 \times 4 \times 4 = 64$ .  
 Number of ways of Correcting 3 Questions =  $1 \times 1 \times 1 = 1$ .  
 So, Required no. of ways =  $64 - 1 = 63$ .

## QUESTION

There are 6 pockets in the coat of a person. In how many ways can he put 4 pens in these pockets?

$$6 \times 6 \times 6 \times 6 = 36 \times 36 = 1296.$$

(Sol)



1

360 ~~X~~

2

1296 ✓

3

4096 ~~X~~

4

None of these ~~X~~

**EXAMPLE**

Find the number of different signals that can be generated by arranging at least 2 flags in order (one below the other) on a vertical staff, if five different flags are available

(Homework Problem)

# EXERCISE – 6.1

## QUESTION

How many 3-digit numbers can be formed from the digits 1, 2, 3, 4 and 5 assuming that

(i) Repetition of the digits is allowed?  $\boxed{5 \mid 5 \mid 5} \rightarrow 5 \times 5 \times 5 = 5^3.$

(ii) Repetition of the digits is not allowed?

$\boxed{5 \mid 4 \mid 3}$

$$\rightarrow 5 \times 4 \times 3 = 60.$$

**Thank**  
**You**